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00001 *****
00002 ** DL Register Area - 120Word
00003 **
00004 ** DW00000 - DW00009 (Pls Use)
00005 ** DW00010 - DW00039 (Bit Use)
00006 ** DW00040 - DW00089 (Word Use)
00007 ** DW00090 - DW00119 (Timer)
00008 *****
00009
00010 "# Program Name      : MTR Cycle 15 (V.S/PreAlignment)
00011 "# Function          :
00012 "# Call Program      : MPS423
00013 "# Sub Program       :
00014 "# Func Program      :
00015
00016 VAR;
00017     LONG MTR_L_CAM_MARK_I           %ML97240;
00018     LONG MTR_R_CAM_MARK_I           %ML97270;
00019
00020     BOOL MTR_ALIGN_L_REQ             %MB970002;
00021     BOOL MTR_ALIGN_R_REQ             %MB970003;
00022
00023     BOOL MTR_L_CAM_LINE_RE           %MB970022;
00024     BOOL MTR_R_CAM_LINE_RE           %MB970023;
00025
00026     BOOL MTR_ALIGN_L_MANUAL_RE        %MB970042;
00027     BOOL MTR_ALIGN_R_MANUAL_RE        %MB970043;
00028
00029     BOOL MTR_L_CAM_DL_ALIGN_RE        %MB970082;
00030     BOOL MTR_R_CAM_DL_ALIGN_RE        %MB970083;
00031
00032     BOOL MTR_ALIGN_L_RESP             %MB970102;
00033     BOOL MTR_ALIGN_R_RESP             %MB970103;
00034
00035     BOOL MTR_L_CAM_LINE_RE           %MB970122;
00036     BOOL MTR_R_CAM_LINE_RE           %MB970123;
00037
00038     BOOL MTR_L_CAM_MANUAL_ALIGN_RE    %MB970142;
00039     BOOL MTR_R_CAM_MANUAL_ALIGN_RE    %MB970143;
00040
00041     BOOL MTR_L_CAM_DL_ALIGN_RE        %MB970182;
00042     BOOL MTR_R_CAM_DL_ALIGN_RE        %MB970183;
00043
00044     BOOL MTR_L_CAM_ALIGN_RESULT_C     %MB970400;
00045     BOOL MTR_R_CAM_ALIGN_RESULT_C     %MB970500;
00046
00047     BOOL MTR_L_CAM_ALIGN_RESULT_I     %MB970401;
00048     BOOL MTR_R_CAM_ALIGN_RESULT_I     %MB970501;
00049
00050     LONG MTR_L_CAM_ALIGN_ERROR_COI    %ML97042;      // 0:OK 1:NO MARK ID 2:MANUAL SEARCH 3:MANUA
00051 L OK 4:MANUAL ABORT
00052     LONG MTR_R_CAM_ALIGN_ERROR_COI    %ML97052;      // 0:OK 1:NO MARK ID 2:MANUAL SEARCH 3:MANUA
00053 L OK 4:MANUAL ABORT
00054
00055     BOOL MTR_L_CAM_MANUAL_ALIGN_IN    %MB970422;
00056     BOOL MTR_R_CAM_MANUAL_ALIGN_IN    %MB970522;
00057
00058     BOOL MTR_L_CAM_MANUAL_ALIGN_ABOI  %MB970424;
00059     BOOL MTR_R_CAM_MANUAL_ALIGN_ABOI  %MB970524;
00060
00061     LONG MTR_L_CAM_ALIGN_SEARCH_TYI    %ML97254;      // 0:Mark Search 1:Edge Search 2:DL 3:Manual
00062 4:Auto Mark Search
00063     LONG MTR_R_CAM_ALIGN_SEARCH_TYI    %ML97284;
00064
00065     BOOL MTR_ALIGN_MANUALSTOP_ALARM    %MB070381;
00066     BOOL MTR_ALIGN_MARK_DISTANCE_ALAI %MB070382;
00067     BOOL MTR_ALIGN_ZERO_VAL_ALARM      %MB070383;
00068
00069     LONG MTR_ALIGN_L_RESULT_XDAI       %ML97044;
00070     LONG MTR_ALIGN_L_RESULT_YDAI       %ML97046;
00071     LONG MTR_ALIGN_R_RESULT_XDAI       %ML97054;
00072     LONG MTR_ALIGN_R_RESULT_YDAI       %ML97056;
00073
00074 END_VAR;
00075 00000 BLK MW24120 MW97244 W8;
00076 00001 BLK MW24120 MW97274 W8;
00077
00078 00002 MTR_L_CAM_MARK_ID = GL49018;
00079 00003 MTR_R_CAM_MARK_ID = GL49018;
00080
00081 00004 CLR DW00050 W20;
00082 00005 ML07394 = 0;      " PreAlign Auto Mark Search X Of
00083
00084 00006 fset Result
00085     ML07398 = 0;      " PreAlign Auto Mark Search Y Of
00086
00087 00007 fset Result
00088     ML07402 = 0;      " PreAlign Auto Mark Search T Of
00089
00090 fset Result

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00082
00083 00008      MTR_ALIGN_L_REQ = 0                                " Alignment Request OFF(Left)
00084 00009      MTR_ALIGN_R_REQ = 0                                " Alignment Request OFF(Right)
00085 00010      MTR_L_CAM_LINE_REQ = 0;
00086 00011      MTR_R_CAM_LINE_REQ = 0;
00087 00012      MTR_ALIGN_L_MANUAL_REQ = 0;
00088 00013      MTR_ALIGN_R_MANUAL_REQ = 0;
00089 00014      MTR_L_CAM_DL_ALIGN_REQ = 0;
00090 00015      MTR_R_CAM_DL_ALIGN_REQ = 0;
00091
00092 00016      IF MW00056 <> 83.                                    " Turn Center Cal

00093 00017          IF !MB071324 == 1;
00094 00018              IF !MB071020 == 1;
00095 00019                  MW07344 = 1;
00096 00020                  MSEE MPS412;
00097 00021                  IEND;
00098
00099 00022          MSEE MPS421;                                            " Alignment Po:
00100 00023          TIM TMW30230;
00101 00024          IEND;
00102 00025      IEND;
00103
00104 00026      IOW MTR_ALIGN_L_RESP | MTR_ALIGN_R_RESP == 0;
00105
00106      // IF !MB000800 == 1                                        " No-Load Operation
00107 00027          DB00020 = 0;
00108 00028          DB00021 = 0;
00109 00029          DW00020 = 0;
00110
00111 00030      WHILE DW00020 < MW30240;
00112 00031          MTR_L_CAM_ALIGN_SEARCH_TYPE = 4;
00113 00032          MTR_R_CAM_ALIGN_SEARCH_TYPE = 4;
00114
00115 00033          MTR_ALIGN_L_REQ = 1                                    " Alignment Request ON(Left

00116 00034          MTR_ALIGN_R_REQ = 1                                    " Alignment Request ON(Right)
00117 00035          IOW MTR_L_CAM_ALIGN_ERROR_CODE == 0;
00118 00036          IOW MTR_R_CAM_ALIGN_ERROR_CODE == 0;
00119
00120 00037          IOW MTR_ALIGN_L_RESP & MTR_ALIGN_R_RESP==          " Alignment Respons ON
00121
00122 00038          IF MTR_L_CAM_ALIGN_RESULT_OK == 1
00123 00039              ML07310 = MTR_ALIGN_L_RESULT_XDATA;                " Alignment Data X1
00124 00040              ML07312 = (MTR_ALIGN_L_RESULT_YDATA * -1)          " Alignment Data Y1
00125 00041              DB00020 = 1;
00126 00042          IEND;
00127 00043          IF MTR_R_CAM_ALIGN_RESULT_OK == 1;
00128 00044              ML07314 = MTR_ALIGN_R_RESULT_XDATA;                " Alignment Data X2
00129 00045              ML07316 = (MTR_ALIGN_R_RESULT_YDATA * -1)          " Alignment Data Y2
00130 00046              DB00021 = 1;
00131 00047          IEND;
00132
00133 00048          MB07360A = MTR_L_CAM_ALIGN_RESULT_NG;
00134 00049          MB07360B = MTR_R_CAM_ALIGN_RESULT_NG;
00135
00136 00050          MTR_ALIGN_L_REQ = 0                                    " Alignment Request OFF(Left)
00137 00051          MTR_ALIGN_R_REQ = 0                                    " Alignment Request OFF(Right)
00138
00139 00052          IF DB00020 & DB00021 == 1;
00140 00053              DW00020 = MW30240 + 1;
00141 00054              IF CW00053 == 1;
00142 00055                  MB07360A = 0;
00143 00056                  MB07360B = 0;
00144 00057              IEND;
00145 00058          ELSE;
00146 00059              DW00020 = DW00020 + 1;
00147 00060              TIM T100;
00148 00061              IEND;
00149 00062          WEND;
00150
00151 00063      IF GL49044 == 0                                            " T Auto Mark Search (0:Use 1:Un
use)
00152 00064          IF (MB07360A | MB07360B) == 1
00153 00065              IF !MB073606 == 1                                " Auto Mark Seach Complete
00154 00066                  MSEE MPS431;
00155 00067                  IEND;
00156 00068              IEND;
00157 00069          IEND;
00158
00159 00070      IF GL49020 == 0                                            " X,Y Auto Mark Search (0:Use 1:
Unuse)
00160 00071          IF (MB07360A | MB07360B) == 1
00161 00072              IF !MB073606 == 1                                " Auto Mark Seach Complete
00162 00073                  MSEE MPS430;
00163 00074                  IEND;
00164 00075          IEND;

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00165 00076      IEND;
00166
00167      " Manual Search Flag
00168 00077      IF MB07360A | MB07360B == 1;
00169
00170 00078          DB000000 = 0;
00171 00079          DB000010 = 0;
00172 00080          DB000011 = 0;
00173
00174 00081      WHILE DB000000 == 0
00175 00082          MTR_L_CAM_ALIGN_SEARCH_TYPE = 0;
00176 00083          MTR_R_CAM_ALIGN_SEARCH_TYPE = 0;
00177
00178 00084          IOW MTR_ALIGN_L_RESP | MTR_ALIGN_R_RESP ==      " Wait Alignment Respons OF
00179 00085          MTR_ALIGN_L_REQ = MB07360A                        " L Camera Manual Alignment Requ
est
00180 00086          MTR_ALIGN_R_REQ = MB07360E                        " R Camera Manual Alignment Requ
est
00181 00087      EOX;
00182
00183 00088      IOW (MTR_L_CAM_MANUAL_ALIGN_ING | MTR_L_CAM_ALIGN_RESULT_OK) & (MTR_R_CAM_MANUAL_ALI
GN_ING | MTR_R_CAM_ALIGN_RESULT_OK) ==      " Alignment Respons O
00184
00185 00089      IF MTR_L_CAM_ALIGN_RESULT_OK & MTR_R_CAM_ALIGN_RESULT_OK == 1;
00186
00187 00090      ELSE;
00188 00091          PFORK Line1 Line2;
00189              Line1:
00190 00092              IF MTR_L_CAM_MANUAL_ALIGN_ING == 1;
00191 00093                  IOW !MTR_L_CAM_MANUAL_ALIGN_ING == 1;
00192 00094                  IOW MTR_ALIGN_L_RESP == 1;
00193 00095              IEND;
00194 00096              JOINTO LabelX1;
00195
00196              Line2:
00197 00097              IF MTR_R_CAM_MANUAL_ALIGN_ING == 1;
00198 00098                  IOW !MTR_R_CAM_MANUAL_ALIGN_ING == 1;
00199 00099                  IOW MTR_ALIGN_R_RESP == 1;
00200 00100              IEND;
00201 00101              JOINTO LabelX1;
00202 00102      LabelX1: PJOINT;
00203
00204 00103      IF MTR_L_CAM_MANUAL_ALIGN_ABORT == 1;
00205 00104          MB000015 = 1;                                " Cycle Stop
00206 00105          MTR_ALIGN_MANUALSTOP_ALARM = 1                " Alignment Resp NG
00207 00106          MTR_ALIGN_L_REQ = 0;
00208 00107          DB000000 = 1;
00209 00108          EOX;
00210 00109      IEND;
00211 00110      IF MTR_R_CAM_MANUAL_ALIGN_ABORT == 1;
00212 00111          MB000015 = 1;                                " Cycle Stop
00213 00112          MTR_ALIGN_MANUALSTOP_ALARM = 1                " Alignment Resp NG
00214 00113          MTR_ALIGN_R_REQ = 0;
00215 00114          DB000000 = 1;
00216 00115          EOX;
00217 00116      IEND;
00218 00117      IEND;
00219
00220 00118      IF MTR_L_CAM_ALIGN_RESULT_OK == 1;
00221 00119          ML07310 = MTR_ALIGN_L_RESULT_XDATA;            " Alignment Data X1
00222 00120          ML07312 = (MTR_ALIGN_L_RESULT_YDATA * -1    " Alignment Data Y1
00223 00121          MB07360A = 0;                                " L Alignment NG (0:OK 1:NG)
00224 00122          MTR_ALIGN_L_MANUAL_REQ = 0                    " L Camera Manual Alignment
Request
00225 00123      IEND;
00226
00227 00124      IF MTR_R_CAM_ALIGN_RESULT_OK == 1;
00228 00125          ML07314 = MTR_ALIGN_R_RESULT_XDATA;            " Alignment Data X2
00229 00126          ML07316 = (MTR_ALIGN_R_RESULT_YDATA * -1    " Alignment Data Y2
00230 00127          MB07360B = 0;                                " R Alignment NG (0:OK 1:NG)
00231 00128          MTR_ALIGN_R_MANUAL_REQ = 0                    " R Camera Manual Alignment
Request
00232 00129      IEND;
00233
00234 00130      MTR_ALIGN_L_REQ = 0                                " L Camera Manual Alignment
Request OFF
00235 00131      MTR_ALIGN_R_REQ = 0                                " R Camera Manual Alignment
Request OFF
00236      //
00237          IOW MTR_ALIGN_L_RESP | MTR_ALIGN_R_RESP ==      " Wait Alignment Respons OFF
00238
00238 00132      IF MB07360A | MB07360B == 0;
00239 00133          DB000000 = 1;
00240 00134      IEND;
00241 00135      EOX;
00242 00136      WEND;
00243 00137      IEND;

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00244 00138      IF !MB000015 == 1                                " Cycle Stop
00245 00139          IF ML07310 == 0                            " Alignment ZeroValue Check
00246 00140              IF ML07312 == 0;
00247 00141                  IF ML07314 == 0
00248 00142                      IF ML07316 == 0;
00249 00143                          MTR_ALIGN_ZERO_VAL_ALARM = 1    " Alignment ZeroValue Alarm
00250 00144                              EOX;
00251 00145                                  IEND;
00252 00146                                      IEND;
00253 00147                                          IEND;
00254 00148      IEND;
00255
00256 00149      DF00050 = IL8516 + ML07310                            " Left Mark Real X Po

00257 00150      DF00052 = IL8596 + ML07314;                        " Right Mark Real X Pos
00258 00151      DF00054 = ML07312 - ML07316;                    " L,R Mark Y Distance
00259 00152      DL00056 = GL49016                                " Align Mark Pos D

00260 00153      DF00058 = DF00052 - DF00050;
00261
00262 00154      DF00060 = (DF00058 * DF00058) + (DF00054 * DF00054);
00263 00155      DF00062 = SQT(DF00060);
00264 00156      DL00064 = DL00056 - DF00062;
00265
00266 00157      IF DL00064 < 0,                                          " Find Mark Distance Check
00267 00158          DL00064 = -DL00064;
00268 00159      IEND;
00269
00270 00160      IF DL00064 > ML30234;
00271 00161          MTR_ALIGN_MARK_DISTANCE_ALARM = 1;
00272 00162      IEND;
00273
00274 00163      IEND;
00275
00276 00164      MSEE MPS405;                                        " Main Transfer 1 Axis Setting
00277
00278      // ELSE;
00279      //      ML07310 = 0;                                        " Alignment Data X1
00280      //      ML07312 = 0;                                        " Alignment Data Y1
00281      //      ML07314 = 0;                                        " Alignment Data X2
00282      //      ML07316 = 0;                                        " Alignment Data Y
00283      // IEND;
00284
00285 00165      RET;

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